

- a) Top b) Bottom
 c) Middle d) Both (a) and (b)
- Q.10 When value of $x_u < x_{u(max)}$ then the section is known as _____
 a) Balanced section b) Under-reinforced section
 c) Over-reinforced section d) None of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Effective cover is taken upto _____ of reinforcement. (Centre/Bottom)
- Q.12 In limit state of collapse partial factor of safety for concrete is _____. (1.0/1.5)
- Q.13 The circular column with helical reinforcement shall be reinforced with at least _____ longitudinal bars. (4Nos./6Nos)
- Q.14 Grade of steel is _____ proportional to shear strength. (Directly/ Inversely)
- Q.15 Single wires used as steel reinforcement are called as _____ (Bond/ Tendons)
- Q.16 The portion of beam consisting of slab is known as _____ (Flange/ Web)
- Q.17 The limiting value of depth of N.A. For Fe-415 steel is _____ (0.45d/ 0.48d)
- Q.18 When the beam is subjected to eccentric loading the singly reinforced beams are provided. (True/False)
- Q.19 When corners of the two-way slab are not held down it is known as _____ (Simply supported/ Cantilever)
- Q.20 In Singly reinforced beams, _____ zone is below the neutral axis. (Compressive zone/ Tensile zone)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write any five salient features of limit state method.
- Q.22 Find the ultimate moment of resistance of a doubly reinforced beam section 250 mm X 450 mm. Compression steel reinforcement = 2 Nos. of 16 mm diameter, Tensile steel reinforcement = 4 Nos. of 25 mm diameter. Effective cover = 50 mm. Use M 25 grade of concrete, Fe 415 grade of steel.
- Q.23 Why stirrups hooks are provided in compression zone?
- Q.24 Why main reinforcement is provided along shorter span in one way slabs?
- Q.25 What are the requirements of a good concrete? (Any five)
- Q.26 Write down the assumptions made in limit state of collapse in flexure.

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- Q.27 A short column 400 mm X 400 mm is reinforced with 4 bars of 20 mm diameter. Find the ultimate load carrying capacity of the column if the minimum eccentricity is less than 0.05 times the lateral dimensions. Use M25 concrete and Fe-415b steel.
- Q.28 Write down the conditions under which the doubly reinforced beams are provided.
- Q.29 Write any five advantages of pre-stresses concrete.
- Q.30 Draw the cross-section of T-beam and write the IS specification for the calculation of effective width of the flange of isolated T-beams.
- Q.31 Write any five advantages of RCC.
- Q.32 A singly reinforced rectangular beam of width 230 mm and 450 mm effective depth is reinforced with 4 bars of 20 mm diameter. Find out the depth of neutral axis and specify the type of beam. Use M25 concrete and Fe-415 steel. Also find the lever arm for the section.
- Q.33 Calculate depth of the slab, effective span for a R.C.C slab simply supported over a room of dimension 4 m x 6m (clear size). The slab carries live load of 4000N/m² and floor finish of 1000 N/m². Take width of bearing as 300 mm. Use M25 grade concrete and Fe 415 steel, the corners of slab not held down.
- Q.34 Write down the assumptions for the limit state of collapse in compression.
- Q.35 Calculate the factored bending moment for a simply supported rectangular beam having clear span 4.5 m and superimposed load 6kN/m. Use M25 concrete and Fe 415 steel.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Design a simply supported RCC one way slab to carry a factored load of 15.75 kN/m² (including self weight) on an effective span of 3.2 m. Bearing on wall = 300 mm. Use M25 concrete and Fe - 415 steel.
- Q.37 Write down the steps for the design of singly reinforced beam using the limit state method of design.
- Q.38 A doubly reinforced beam is 200 mm wide and 350 mm deep to the center of the tensile reinforcement. The areas of the compression and tensile steel are 1245 mm² and 1600 mm² respectively. The effective cover to the compression reinforcement is 50mm. Find the ultimate moment of resistance of the beam section. Use M25 concrete and Fe-415 steel.

SECTION-E

Note: Long answer type questions. Attempt any two questions out of three questions. (2x25=50)

- Q.39 Draw a suitable scale sectional plan and sectional elevation of a square column with an isolated footing of uniform thickness from the following data:

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